



DAX Information Functions

This A4-ready document is a concise, print-friendly reference for Power BI analysts and BI developers covering DAX Information Functions: definitions, syntax, parameters, return types, targeted examples, and practical use cases. Use this guide for calculated columns, measures, data validation, and model debugging.

Bythor. Fubriaish

Introduction

Information functions in DAX are predicate and descriptive functions that test values or return information

about the type or state of a value. They are most often used to:

- Validate and sanitize incoming or transformed data (data quality checks).
- Create conditional logic for calculated columns and measures.
- Prevent runtime errors by detecting blanks, errors, or unexpected types.

This section explains the typical scenarios: conditional

formatting, defensive coding in measures, pre-

aggregation validation, and runtime checks for ETL/Power Query outputs integrated into the model.

Overview Table for Quick Reference

FunctionName	Description	Return Type	Category
CONTAINS()	Tests whether a table contains specified rows matching criteria	TRUE/FALSE	Table / Logical
CONTAINSROW()	Checks whether a table contains a single row of specific values	TRUE/FALSE	Table / Logical
ISBLANK()	Returns TRUE if value is blank (DAX Blank)	TRUE/FALSE	Information
ISERROR()	Returns TRUE if expression results in an error	TRUE/FALSE	Information
ISEVEN()	Returns TRUE if a number is an even integer	TRUE/FALSE	Numeric
ISEVEN()	Checks whether value is TRUE or FALSE	TRUE/FALSE	
ISEVEN()	Returns TRUE if value is not text	TRUE/FALSE	
ISLOGICAL()		TRUE/FALSE	Information
ISLOGICAL()		TRUE/FALSE	Information
ISNONTEXT()		TRUE/FALSE	Information
ISNONTEXT()		TRUE/FALSE	Information
ISNUMBER()	Returns TRUE if value is numeric	Table	Numeric
ISODD()	Returns TRUE if a number is an odd integer		
ISTEXT()	Returns TRUE if value is text		Information
UNION()	Combines two tables with identical column structure (optional reference)		Table / Transform

Use this table as a compact lookup. Later sections provide deeper syntax, parameters, examples, and recommended usage patterns.

CONTAINS() and CONTAINSROW() 4

Table membership tests

Definition

`CONTAINS(table, columnName, value[,columnName2, value2 ...])` returns TRUE when any row in table

matches all column/value pairs. `CONTAINSROW(table, value1 [, value2 ...])` compares a full row (no column names) against the table.

Syntax & Parameters

- `CONTAINS(table, column, value, ...)`: table 3 table expression; column 3 column reference; value 3 scalar expression to compare.
- `CONTAINSROW(table, value1, value2...)`: values must match table column order.

Example

`CONTAINS(Products, Products[ProductKey], 12345)` -- returns TRUE if ProductKey 12345 exists.

Use Case

Use in measures for membership checks (e.g., "Is this SKU in the promotion list?") 4 more efficient than `FILTER+COUNTRROWS` when checking existence.

ISBLANK() and ISERROR() 4 Detecting missing or faulty values

Definition & Syntax

ISBLANK(value) 4TRUEwhenvalue is DAXBLANK(propagatedfromnull/empty).ISERROR(value)4TRUE when expression evaluation produces an error (divide by zero, type mismatch).

Examples

Measure example: SafeAvg := IF(ISBLANK([Count]), BLANK(), [Sum] / [Count]) Error detection: HasError := ISERROR([PotentialErrorExpression])

Notes & Pitfalls

- ISBLANK differs from comparing to "" 4 text empty string is not BLANK; use LEN or ISTEXT to disambiguate.
- ISERROR can be used to guard complex expressions; prefer TRY... style defensive logic when available in transformation stages.

Type-checking Functions 4 ISNUMBER, ISTEXT, ISLOGICAL, ISNONTEXT

These functions return TRUE when a value is of the specified DAX type. Use them to enforce type expectations and drive conditional logic.



ISNUMBER()

Returns TRUE for numeric values. Example:
`IF(ISNUMBER([Column]), [Column],
BLANK())`.



ISTEXT()

Returns TRUE for text values. Useful before text transformations like SEARCH or CONCATENATE.



ISLOGICAL()

Detects boolean TRUE/FALSE. Helpful when booleans may be imported as integers or text.



ISNONTEXT()

TRUE if not text; use to catch numbers, blanks, and booleans before text-only operations.

Example combined pattern: `ValidNumeric := IF(AND(NOT ISBLANK([Value]), ISNUMBER([Value])), [Value], BLANK())`

ISEVEN() and ISODD() 4 Integer parity checks

ISEVEN(number) and ISODD(number) return TRUE when value is an even or odd integer respectively. They

implicitly convert to integer if the value is numeric; non-numeric values return error or FALSE depending on context.

Syntax & Example

PromoGroup := IF(ISEVEN([OrderID]), "Even Batch", "Odd Batch") Ensure you validate with ISNUMBER before calling parity functions to avoid unexpected errors.

Use Cases

- Partitioning rows for A/B style sampling without extra columns.
- Assigning alternating formats in visuals when a natural row order is present.

UNION() for Optional reference for table combination

UNION(table1,table2) returns a table that appends rows from table2 to table1. Columns are matched by position; identical structure is recommended. While not an 'Information' predicate function, it frequently appears in examples where information checks follow table concatenation.

Example

CombinedPromotions := UNION(CurrentPromotions, HistoricalPromotions) Then use CONTAINS(CombinedPromotions, [PromoID], targetID) to test membership.

Notes

- UNION does not remove duplicates 4 use DISTINCT or EXCEPT for deduplication.
- Column order matters; prefer SELECT COLUMNS to ensure consistent schema before UNION.

Practical Patterns, Performance, and Pitfalls

Common Patterns

- Defensivemeasures: GuardwithISBLANK/ISNUMBERtoavoidDIV/0 andtype errors.
- Membership checks: Use CONTAINS when checking existence to avoid scanning with FILTER/COUNTROWS.
- Type normalization: Convert or coerce types in Power Query when possible; use IS* functions for last-mile checks.

Performance Considerations

Prefer table-level functions that short-circuit (CONTAINS) over row-by-row FILTER when possible. Move heavy transforms to Query Editor. Use IS functions sparingly in large row context calculations that iterate per row without necessary filter propagation.

Common Pitfalls

- 1 Confusing BLANK() with empty string 4 test with ISBLANK and ISTEXT/LEN as needed.
- Calling parity functions on non-integers without validation.
- 2 Relying on UNION without aligning schemas causing silent mismatches.
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Summary, Tips, and Quick Cheat Sheet

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Functions Covered

CONTAINS, CONTAINSROW, ISBLANK, ISERROR, ISEVEN, ISODD, ISLOGICAL, ISNONTTEXT, ISNUMBER, ISTEXT, UNION

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Quick Rules

Validate types before operations.
Use table-level existence checks.
Prefer Query Editor for heavy coercion.

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Common Returns

Most information functions return TRUE/FALSE. Use in IF, FILTER, or to guard calculations.

Final recommendations: standardize types at the ETL layer, use IS functions for lightweight guards, prefer CONTAINS for existence checks, and document assumptions in the model. For more details and the latest behavior, consult Microsoft Learn DAX reference.

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